THE FLEET — ONE NETWORK, MANY SHAPES

NODE	NAME	ROLE	ENVELOPE	POWER	STATUS
L1	Brick	field node — mailbox+repeater+bulletin+SOS	160×80×24 mm	4×18650/solar	CAD ready
E	Totem	community board — public E-Ink + mesh node	~320×220×80+pole	solar+LiFePO4	spec
S	Mite	swarm unit — perchable env sensor	~90×90×35 mm	solar+LiPo	CAD scripted
A	Anchor	charging station — dock+relay+backhaul	~350×250×400+array	20–60 W+LiFePO4	spec
P	ArcticPhone	rural comms — mesh-native, sat SOS	~145×72×14 mm	replaceable+solar	concept
D	Seed	degradable mote — seasonal sensing	~20×20×5 mm	printed+harvest	R&D
H	Meridian	satellite payload — mesh IS ground segment	1U/3U CubeSat	solar array	vision

SHARED DESIGN DNA (EVERY NODE)

— Climate: -45..+70 °C · IP65+ · UV-stable · LiFePO4-preferred · December-sized solar.
— Power: solar-first · 30–90 day autonomy · works with zero grid.
— Field: bear-country mounts · two-person install · hand-tool repair.
— Sovereignty: community-owned mesh · data stays home · OCAP · EN/FR · privacy by design.
— Appropriate tech: right-sized cost · repairable > sealed · open docs.
— Offline dignity: works when everything else has failed — that IS the product.

DRAWING INDEX

AF-000 this sheet — fleet overview & index
AF-L1-101…103 L1 Brick — arrangement, parts/power, assembly
AF-S-101…103 S Mite — arrangement, parts/power, assembly/roles
AF-E-101…102 E Totem — arrangement, parts/install
AF-A-101…102 A Anchor — arrangement, parts/power
AF-P-101…102 P ArcticPhone — concept + spec sketch
AF-D-101 D Seed — R&D brief (degradable motes)
AF-H-101 H Meridian — satellite mission concept

HOW TO READ THIS SET

— L1 Brick sheets are build-ready — real BOM, wiring, assembly, acceptance test.
— S Mite sheets are prototype-ready — orderable parts, CAD scripted (cad/swarm_node.py).
— E Totem / A Anchor sheets are spec-level — real parts selected, needs CAD pass.
— P Phone / D Seed / H Meridian are concept/R&D — honest status on every sheet.
— Every sheet: SURAFEL NEGUSSIE drew it · EMEBET INC. owns it · PROPRIETARY & CONFIDENTIAL.

ARCTICCHAIN — RURAL NODE FLEET

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